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REVIEW

Changes in axial length after orthokeratology lens treatment for myopia: a meta-analysis

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Abstract
Purpose Orthokeratology (OK) lens is a popular optical method to control myopia progression. This study aimed to assess the effect of OK lens on axial length change compared with glasses.
Methods PubMed, Web of Science, and Embase were searched to retrieve the related articles. Then, the articles were selected according to predefined criteria. Standardized mean difference (SMD) and 95% confidence interval (CI) were selected as effect size for combining and analyzing the change in axial length.
Results A total of 13 articles were included in the present study. Different models were selected according to the heterogeneity of each analysis. The axial length change in OKs group was significantly smaller than control group; SMD (95% CI) of change in axial length was − 0.857 (− 1.146, − 0.568), $p < 0.001$ at the end of 1 year and − 0.701 (− 1.675, 0.272), $p < 0.001$ at the end of 2 years or longer time.
Conclusions OK lens treatment appears more effective in slowing axial elongation than glasses during the early treatment of myopia in children.

Keywords Myopia · OK lens · Axial length · Change · Meta-analysis

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Introduction

Myopia, known as short-sightedness or near-sightedness, is one of the most common eye disorders. It begins in early life and the frequency and severity get evolved through childhood and adolescence into adulthood [1]. According to the data published in World Health Organization (WHO) and International Agency for the Prevention of Blindness (IAPB), myopia might be the main cause of visual impairment, which has become a severe public health problem [2, 3]. As reported, the prevalence of myopia has increased worldwide [4–6]. The prevalence of myopia in America increased 66% over a 30-year period ending in 2004 [7]. The prevalence in Asian populations approaches the rate of 37–60% [8, 9]. In East Asia, especially in China, 80–90% students suffered from myopia since childhood [2]. As the most popular eye disorder, myopia is responsible for the increased healthcare costs and a number of degenerative eye diseases including macular degeneration, cataract, and glaucoma [10–12]. It is urgent to correct and control the children's myopia [13]. Different methods are used to slow the progression of myopia in children, including atropine, rigid contact lenses, and physical exercise [14, 15]. In recent years, orthokeratology (OK) lens has become more and more prevalent in myopia treatment.

By special design and mechanical compression, OK lens can change the shape of cornea and improve eyesight. Using OK lens not only slows down the progression of myopia but also makes people more convenient in daily life by providing a good naked vision. A number of primary studies have been published to analyze and concluded that OK lens can slow the progression of myopia in children effectively [16–18]. However, few studies focused on the accommodative response including the changes of axial length after receiving OK lens treatment, which was closely related to the refractive status of cornea and visual quality. In this study, we compared the impact on axial length changes between glasses, the very traditional treatment for myopia, and OK lens treatment to assess the advantage of OK lens in myopia treatment of children.

Materials and methods

Literature research and screening

"Myopia," "orthokeratology (OK) lens," and "axial length" were used as keywords, and databases including PubMed, Web of Science, and Embase were searched to retrieve the related articles up to December 2017. Manual search was also conducted as a complementary method by reviewing the reference lists and prospective citation search for all retrieved studies, and relevant review articles were finished at the meantime. Then, the titles, abstracts, and full texts of included articles were screened step by step according to inclusion and exclusion criteria. All of the literature research and screening were conducted by two independent researchers. Any disagreements were resolved by discussion with a third researcher. All related articles were included if they met the following criteria: (1) Study samples were patients with acquired myopia without organic lesion and younger than 18 years old; (2) Studies which focused on the effect on axial length by orthokeratology lens treatment or ordinary glasses treatment; (3) Information including axial length measurements, myopic refraction, astigmatism degree, and raw data was available; (4) Intraocular lens master optical biometer (IOL Master) was used for axial length measurement; (5) English written articles. Studies were excluded if they were: (1) Articles with duplicate data; (2) Abstract, comment, review and editorial reviews or case reports; (3) Study samples with other eye diseases, like cataracts, glaucoma, and amblyopia; (4) Studies with no sufficient data, or in which relevant raw data could not be abstracted.

Research quality assessments

The methodological quality assessment of the included studies was conducted by two independent researchers according to the guidelines of the Newcastle–Ottawa Scale (NOS). This scale uses a star system (with a maximum of nine stars), which is composed of three domains [19]: selection of participants (including four items), comparability of study groups (including two items), and the ascertainment of exposure or outcome (including three items), which are used to judge the applicability and risk of bias [20]. Each question was answered with "yes," "no," or "unclear." A "yes" answer represents the low risk of bias and is assigned a score of 1, while a "no" or "unclear" answer represents a high risk of bias and is assigned a score of 0. Thus, the quality score of study ranged from 0 to 9 stars. Higher scores were indicative of lower risk of bias [20]. Studies that scored 7 stars or greater than 7 stars were regarded as lower risk of bias, and those that scored less than 7 stars indicated higher or moderate risk. All the NOS of the included studies are shown in Table 1.

Data extraction

General information containing the publication year, sample area, and study type together with clinical data like follow-up time, sample number, age, axial length measurements, myopic refraction, and astigmatism degree was extracted. The data extractions were also conducted by two independent researchers. Any disagreements were resolved by discussion with a

Table 1 Characteristics of the included studies

References	Country	Duration	Study design	Random trial	Follow-up time	Group	Number	Age	Myopia	Astigmatism	Axial length measurements	NOS
Swarbrick [6]	East Asia	–	Prospective	Yes	1 year	OKs/GP	26/26	10.08–17	− 1.00 to − 4.00D	< 1.5D	IOL Master ocular biometer (Zeiss, Jena, Germany)	9
Zhu [13]	China	2008–2009	Retrospective study	Yes	2 years	OKs/spectacles	65/63	7–14	> −1.25D	< 1.50D	IOL Master (Carl Zeiss Jena GmbH, Jena, Germany)	7
Hiraoka [17]	Japan	2002–2010	Retrospective study	–	5 years	OKs/spectacles	22/21	8–12	− 0.5 to − 5.00D	< 1.50D	IOL Master (Carl Zeiss Meditec, Dublin, CA, USA)	10
Cho [21]	China	–	Prospective	Yes	3 years and 2 months	OKs/SV	16/13	3–13	–	< 0.25D	IOL Master (Zeiss Humphrey Dublin. CA, USA)	9
Chen [22]	China	–	–	No	2 years	OKs/SV	35/23	6–12	0.5 to 5.00D	< 1.50D	Zeiss IOL Master (Zeiss Humphrey Systems, Dublin, CA, USA)	8
Santodomingo-Rubido [23]	Spain	2007–2008	Prospective	No	7 years	OKs/SV	31/30	6–12	− 0.75 to − 4.00D	< 1.00D	Zeiss IOL Master (Carl Zeiss Jena GmbH, Jena, Germany)	8
Charm [24]	China	–	–	Yes	2 years	OKs/spectacles	26/26	8–11	> −5.00D	< 2.00D	Zeiss IOL Master (Carl Zeiss Meditec, Inc.)	8
He [25]	China	2013–2014	Retrospective study	No	1 year	OKs/SV	141/130	7.0–11.5	− 0.50 to − 6.00D	< 1.5D	IOL Master (IOL: intraocular lens; Carl Zeiss Meditec AG, Jena, Germany)	
Cho [26]	China	2008–2009	–	Yes	1 year	OKs/SV	37/41	6–10	− 0.5 to − 4.00D	< 1.25D	IOL Master (Zeiss Humphrey, Dublin, CA, USA)	9
Cheung [27]	China	–	–	Yes	2 years	OKs/SV	37/38	7–10	− 0.5 to − 4.00D	< −1.250D	IOL Master (Zeiss Humphrey Dublin, CA, USA)	
Kakita [28]	Japan	2002–2007	Prospective	–	2 years	OKs/spectacles	42/50	8–16	− 0.5 to − 10.00D	< 1.5D	IOL Master (Carl Zeiss Meditec, Dublin, CA, USA)	9
Pauné [29]	Spain	2011–2012	Prospective	No	2 years	OKs/SV	29/41	9–16	− 0.75 to − 7.00D	< −1.25D	–	9
Na [30]	Korea	2013–2016	Retrospective study	No	1 year	OKs/without OKs	45/45	7–13	− 0.75 to − 4.25D	< 1.5D	AL-scan optical biometer (Nidek)	8

OKs group wearing orthokeratology lens, *SV* group wearing single-vision spectacles, *D* diopter, *NOS* Newcastle–Ottawa Scale third researcher. Raw data would be extracted if we could contact primary investigators of these studies.

Statistical analysis

Due to the different measurements, standardized mean difference (SMD) and 95% confidence interval (CI) were chosen as effect size to combine and analyze the change in axial length. Cochran's Q test and I^2 test were carried out to analyze the heterogeneity among the included articles. A P value of Cochran's Q test above 0.05 or $I^2 > 50\%$ indicated moderate or high levels of heterogeneity and random effect model would be utilized to accomplish this meta-analysis. On the contrary, fixed effect model would be used. Begg's test ($p = 0.131$) and Egger's test were used to assess publication bias. To assess the sensitivity of this meta-analysis, we compared the merged results with all studies and the merged results by removing each study. All these statistical analyses were carried out by STATA 12.0 software.

Results

Search results

As shown in Fig. 1, a total of 757 articles were searched using index words from three databases (PubMed: 343, Web of science: 52, Embase: 362). After removing 292 duplicated articles and seven articles with no titles and authors in endnote software or manually, 458 articles were screened by reading titles and abstracts. Fifty-eight articles with only abstract or conference articles, 109 reviews, 36 case reports, 27 articles not written in English, and 168 articles with a single method or the treatment changed were excluded. Then, the remaining 60 articles were full text screened. According to the inclusion and exclusion criteria, studies with repeated data, subjects of comparison in studies wear no glasses, the same person with two eyes of different treatments, or the samples older than 18 years old were excluded. Though Cho P., Charm J., and Cheung SW were the

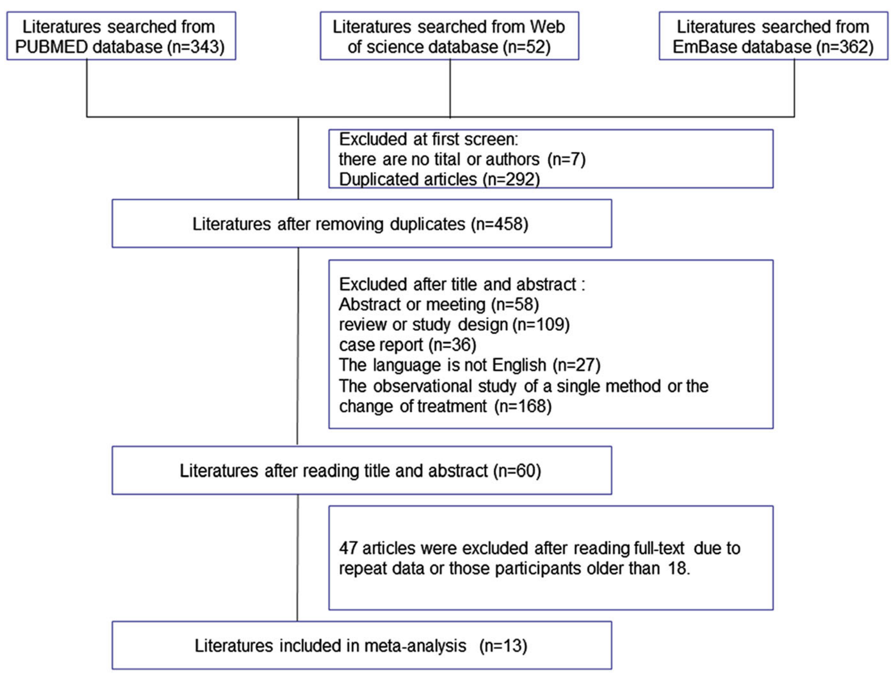


Fig. 1 Literature screening progress

members of the same experimental group, different myopia patients were recruited during their researches, respectively. Therefore, there was no duplicate data among these studies. Finally, 13 studies [6, 13, 17, 21–30] were included in this meta-analysis.

As given in Table 1, the included 13 studies were checked by two independent researchers according to the NOS guidelines. After checking the three domains (selection of participants, comparability of study groups, and the ascertainment of exposure or outcome), all the 13 studies got seven or greater scores. Two of the 13 studies were based on Spain and 11 studies concentrated on samples from East Asia including Japan, Korea, and China. The characteristics of these studies are shown in Table 1. All myopia patients from these studies were recruited at an early age, and followed up at least 1 year. The degree of myopia ranged from − 0.5D to − 10.0D among recruited children, and the degree of astigmatism was all under 2.00D. In terms of the axial length, all recruited studies chose IOL Master as the main measurement to ensure the accuracy. The axial length increment at different time points of each included article is given in Table 2.

Effects of intervention

Change in axial length at the end of 1 year

In this part, we analyzed the change in axial length at the end of 1 year among children from 6 to 12 years old. A total of 408 children received OKs treatment and 396 children as control of seven studies were analyzed. According to the heterogeneity results ($I^2 > 50\%$, $p < 0.05$), random effect model was chosen to get the merged results. As shown in Fig. 2a, SMD (95% CI) of change in axial length at the end of 1 year was − 0.857 (− 1.146, − 0.568), $p < 0.001$; the variation in axial length in OKs group was significantly smaller than control group after one year's treatment, indicating that OKs treatment appears effective in controlling the axial extension in the first year. According to Begg's test ($p = 0.592$) and Egger's test ($p = 0.618$), together with the symmetrical plots in funnel plot (Fig. 2b), no publication bias was found in this meta-analysis. In addition, low sensitivity suggested a stable and reliable merged result (Fig. 2c).

Change in axial length at the end of 2 years or longer time

A total of 303 children received OKs treatment, and 308 children as control of seven studies were analyzed. Heterogeneity analysis was carried out first. Due to I^2 value (50.4%) and the p value of Q test (0.041), the random effect model was chosen. At the end of 2 years, the merged SMD (95% CI) was − 0.701 (− 1.675, 0.272), $p < 0.001$ (Fig. 3a), which was consistent with the merged result of change in axial length after 1 year. The axial length change in OKs group was also significantly smaller than control group. Publication bias was carried out by both Begg's test ($p = 0.466$) and Egger's test ($p = 0.010$). Symmetrical funnel plot (Fig. 3b) showed that little publication bias was found in this meta-analysis. Also, the sensitivity analysis indicated that this merged SMD was very stable (Fig. 3c).

Discussion

As the most popular eye disorder, the prevalence of myopia all over the world turns to keep increasing by the year of 2050 [31]. It has been proved that myopia is a consequence of uncoordinated contributions of ocular components to overall eye structures, and the cornea and lens failed to compensate for axial length elongation [32]. Along with the progression of myopia, eyeball would grow abnormally, a longer axial, and the image of a distant object falls in front of the photoreceptors, and cannot be brought into focus by accommodation [2]. Thus, the measurement of axial length is widely evaluated in myopia studies [33]. Studies showed that people turned to suffer with myopia since childhood in recent decades. In the meantime, various kinds of methods were applied to control the progression of myopia for children. The interventions to reduce myopia mainly included four kinds of methods: outdoor intervention, optical intervention, pharmacological intervention, and scleral reinforcement [2, 34–36]. Due to the instant effect and convenient operation, optical treatment has been the principal method to correct and control myopia. OK lens, one of the optical interventions, can correct refractive errors using custom-designed reverse geometry contact lenses to temporarily modify cornea curvature [37–39].

Table 2 Outcomes of the included studies

References	Axial length (mm)							
	Baseline		12 months		24 months		≥ 36 months	
	Control	OKs	Control	OKs	Control	OKs	Control	OKs
Swarbrick [6]	–	–	0.09 ± 0.12*	− 0.05 ± 0.11*	–	–	–	–
Zhu [13]	24.85 ± 1.08	24.91 ± 0.83	25.25 ± 1.06	25.07 ± 0.82	25.56 ± 1.02	25.26 ± 0.87	–	–
Hiraoka [17]	24.22 ± 0.71	24.09 ± 0.77	0.38 ± 0.20*	0.19 ± 0.09*	0.33 ± 0.18*	0.26 ± 0.13*	0.29 ± 0.16*	0.19 ± 0.15*
Cho [21]	24.07 ± 0.79	24.26 ± 0.80	–	–	–	–	24.69 ± 0.88	24.72 ± 0.90
Chen [22]	24.18 ± 1.00	24.37 ± 0.88	0.36 ± 0.16*	0.15 ± 0.18*	0.64 ± 0.31*	0.31 ± 0.27*	–	–
Santodomingo-Rubido [23]	24.08 ± 0.27	24.39 ± 0.23	24.46 ± 0.27	24.41 ± 0.23	24.78 ± 0.26	24.81 ± 0.25	25.43 ± 0.27	25.30 ± 0.31
Charm [24]	25.97 ± 0.53	26.05 ± 0.80	0.30 ± 0.19*	0.06 ± 0.11*	0.51 ± 0.32*	0.19 ± 0.21*	–	–
He [25]	24.82 ± 0.77	24.71 ± 0.72	25.20 ± 0.78	24.98 ± 0.70	–	–	–	–
Cho [26]	24.40 ± 0.84	24.48 ± 0.71	0.37 ± 0.16*	0.20 ± 0.15*	0.63 ± 0.26*	0.36 ± 0.24*	–	–
Cheung [27]	24.37 ± 0.84	24.52 ± 0.71	25.75 ± 0.90	24.72 ± 0.70	25.02 ± 0.90	24.88 ± 0.72	–	–
Kakita [28]	24.79 ± 0.80	24.66 ± 1.11	–	–	0.54 ± 0.27*	0.29 ± 0.27*	–	–
Pauné [29]	24.36 ± 0.81	24.77 ± 0.89	0.28 ± 0.17*	0.15 ± 0.10*	0.52 ± 0.22*	0.32 ± 0.20*		
Na [30]	23.50 ± 0.68	24.19 ± 0.82	23.82 ± 0.70	24.26 ± 0.79	–	–	–	–

OKs group wearing orthokeratology lens

*Mean ± SD of axial length increment at the time point

Fig. 2 Analysis of change in axial length at the end of 1 year

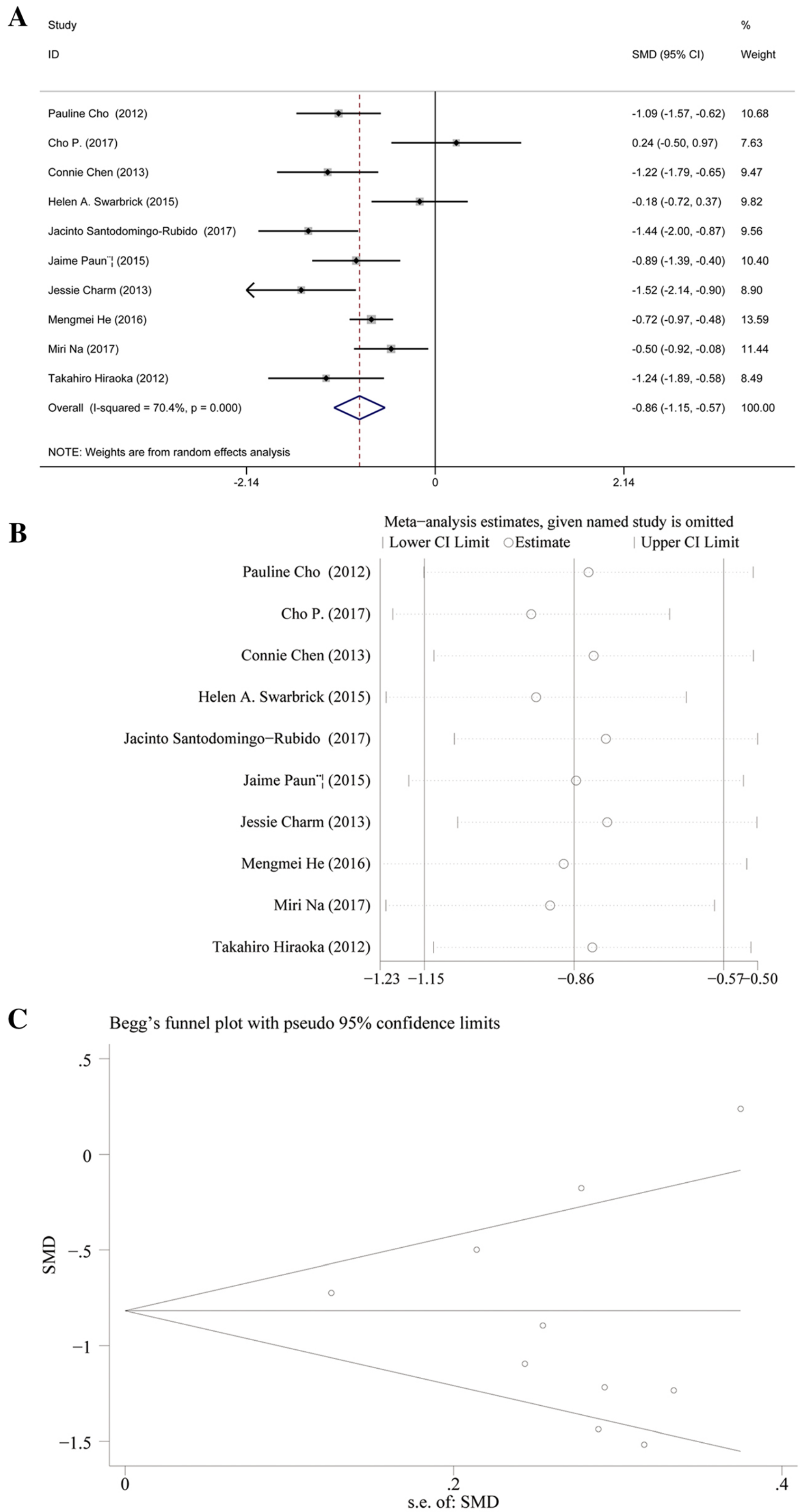
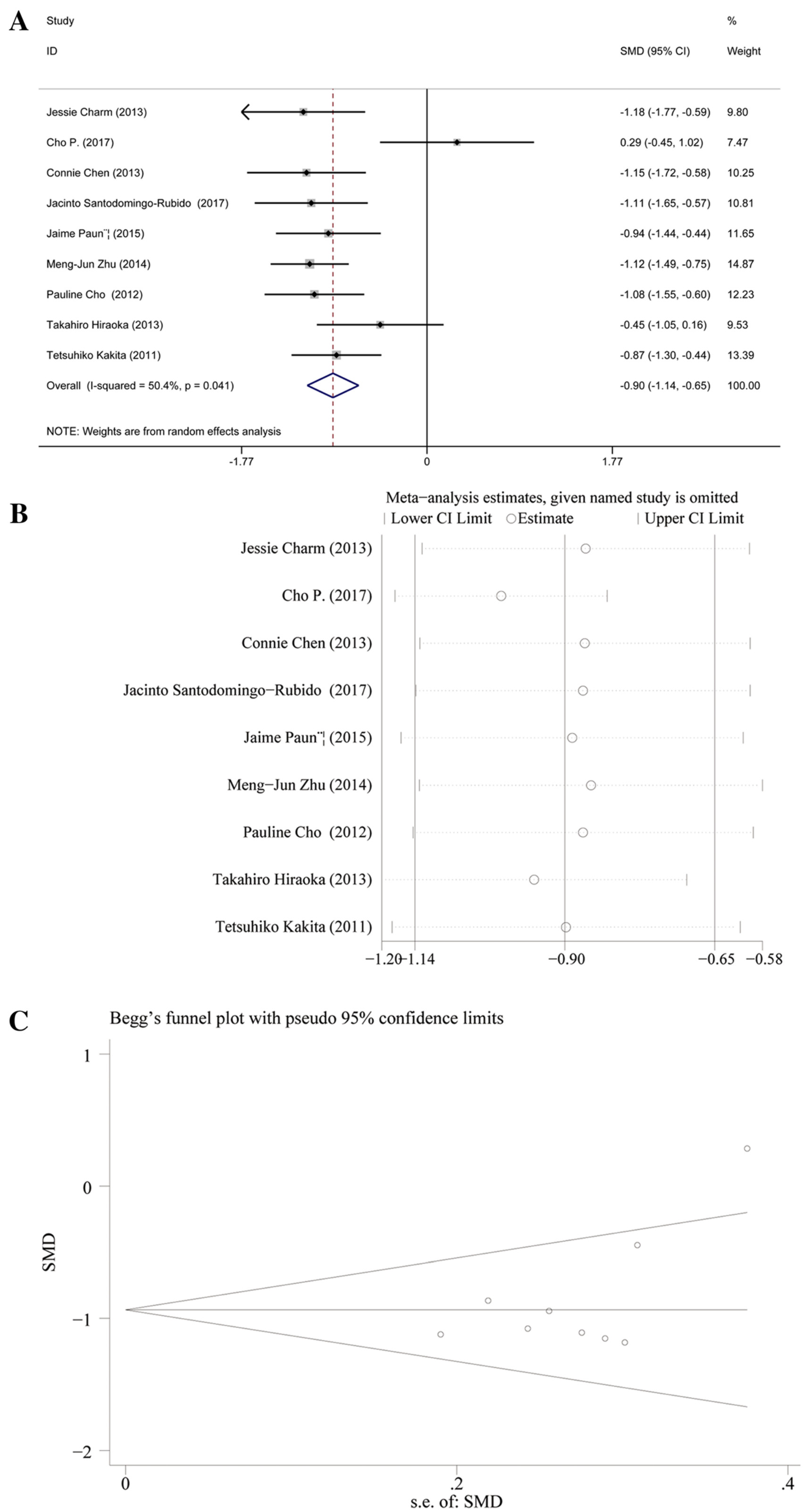


Fig. 3 Analysis of change in axial length at the end of 2 years or longer time



Though no direct evidence has been concluded about the effect of OK lens treatment on children after dozens of studies, researchers found some relationships between OK lens and change of axial length during myopia progression. Cheung et al. [40] reported a case that axial elongation was less in the left eye than in the right after 2 years, which indicated that myopia progression has been slowed down by OK lens wear in a short-sighted eye of a boy who received unilateral OK lens treatment. More studies were done in the recent decades, and it has been reported by Swarbrick [39], Si [41], Hiraoka [17, 28], and lots of other researchers [16, 42, 43] that OK lens treatment could significantly inhibit myopia progression in children by slowing axial length. In our meta-analysis, the data from 13 studies of OK lens effect on slowing axial elongation compared with single glass in myopic children aged from 3–16 years were analyzed. The results suggested that OK lens was more effective in slowing axial elongation than control group at the end of 1 year or 2 years after treatment, especially at the end of 1 year among children ranged from 6 to 12 years old, which was consistent with those studies mentioned above. However, the long-term results showed that there was no statistical difference between these two treatments. The reason for the negative results might be there were little studies with data of axial length change at the end of longer time in this study. Furthermore, heterogeneity should be taken into consideration though no publication bias existed.

In addition, some limitations of this meta-analysis should be noted. First, only data of axial length change were merged to assess the therapeutic effect of OK lens. More items should be investigated to explore the accommodative response of eye balls after receiving OK lens treatment. Only three of the included studies have done 3-year or longer follow-up researches; more studies with longer follow-up period are required to investigate the long-term consequences of OK lens treatment accurately. Third, of the included 13 studies, only two studies were from Spain, others were all from East Asia. Also, we excluded single-vision soft lenses as control which might reduce the number of papers. More studies should be done to find out OK lens's advantages.

Conclusion

In summary, OK lens treatment is more effective than wearing normal glasses to slow axial elongation during the early treatment of myopia in children. More studies with long-term follow-up data are expected to draw a precise conclusion for myopia treatment.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

References

1. Rudnicka AR, Kapetanakis VV, Wathern AK, Logan NS, Gilmartin B, Whincup PH, Cook DG, Owen CG (2016) Global variations and time trends in the prevalence of childhood myopia, a systematic review and quantitative meta-analysis: implications for aetiology and early prevention. Br J Ophthalmol 100(7):882–890. https://doi.org/10.1136/bjophthalmol-2015-307724
2. Morgan IG, Ohno-Matsui K, Saw SM (2012) Myopia. Lancet 379(9827):1739–1748. https://doi.org/10.1016/S0140-6736(12)60272-4
3. Resnikoff S, Pascolini D, Mariotti SP, Pokharel GP (2008) Global magnitude of visual impairment caused by uncorrected refractive errors in 2004. Bull World Health Organ 86(1):63–70
4. Morgan I, Rose K (2005) How genetic is school myopia? Prog Retin Eye Res 24(1):1–38. https://doi.org/10.1016/j.preteyeres.2004.06.004
5. Pan CW, Ramamurthy D, Saw SM (2012) Worldwide prevalence and risk factors for myopia. Ophthalmic Physiol Opt 32(1):3–16. https://doi.org/10.1111/j.1475-1313.2011.00884.x
6. Swarbrick HA, Alharbi A, Watt K, Lum E, Kang P (2015) Myopia control during orthokeratology lens wear in children using a novel study design. Ophthalmology 122(3):620–630. https://doi.org/10.1016/j.ophtha.2014.09.028
7. Vitale S, Sperduto RD, Ferris FL (2009) Increased prevalence of myopia in the United States between 1971–1972 and 1999–2004. Arch Ophthalmol 127(12):1632–1639. https://doi.org/10.1001/archophthalmol.2009.303
8. Saw SM, Goh PP, Cheng A, Shankar A, Tan DT, Ellwein LB (2006) Ethnicity-specific prevalences of refractive errors vary in Asian children in neighbouring Malaysia and

Singapore. Br J Ophthalmol 90(10):1230–1235. https://doi.org/10.1136/bjo.2006.093450
9. Congdon N, Wang Y, Song Y, Choi K, Zhang M, Zhou Z, Xie Z, Li L, Liu X, Sharma A, Wu B, Lam DS (2008) Visual disability, visual function, and myopia among rural chinese secondary school children: the Xichang Pediatric Refractive Error Study (X-PRES)–report 1. Invest Ophthalmol Vis Sci 49(7):2888–2894. https://doi.org/10.1167/iovs.07-1160
10. Saw SM, Gazzard G, Shih-Yen EC, Chua WH (2005) Myopia and associated pathological complications. Ophthalmic Physiol Opt 25(5):381–391. https://doi.org/10.1111/j.1475-1313.2005.00298.x
11. Rein DB, Zhang P, Wirth KE, Lee PP, Hoerger TJ, McCall N, Klein R, Tielsch JM, Vijan S, Saaddine J (2006) The economic burden of major adult visual disorders in the United States. Arch Ophthalmol 124(12):1754–1760. https://doi.org/10.1001/archopht.124.12.1754
12. Verhoeven VJ, Wong KT, Buitendijk GH, Hofman A, Vingerling JR, Klaver CC (2015) Visual consequences of refractive errors in the general population. Ophthalmology 122(1):101–109. https://doi.org/10.1016/j.ophtha.2014.07.030
13. Zhu MJ, Feng HY, He XG, Zou HD, Zhu JF (2014) The control effect of orthokeratology on axial length elongation in Chinese children with myopia. BMC Ophthalmol 14:141. https://doi.org/10.1186/1471-2415-14-141
14. Smith EL III (2013) Optical treatment strategies to slow myopia progression: effects of the visual extent of the optical treatment zone. Exp Eye Res 114:77–88. https://doi.org/10.1016/j.exer.2012.11.019
15. Lee JJ, Fang PC, Yang IH, Chen CH, Lin PW, Lin SA, Kuo HK, Wu PC (2006) Prevention of myopia progression with 0.05% atropine solution. J Ocul Pharmacol Ther 22(1):41–46. https://doi.org/10.1089/jop.2006.22.41
16. Walline JJ, Jones LA, Sinnott LT (2009) Corneal reshaping and myopia progression. Br J Ophthalmol 93(9):1181–1185. https://doi.org/10.1136/bjo.2008.151365
17. Hiraoka T, Kakita T, Okamoto F, Takahashi H, Oshika T (2012) Long-term effect of overnight orthokeratology on axial length elongation in childhood myopia: a 5-year follow-up study. Invest Ophthalmol Vis Sci 53(7):3913–3919. https://doi.org/10.1167/iovs.11-8453
18. Sun Y, Xu F, Zhang T, Liu M, Wang D, Chen Y, Liu Q (2015) Orthokeratology to control myopia progression: a meta-analysis. PLoS ONE 10(4):e0124535. https://doi.org/10.1371/journal.pone.0124535
19. Maddern GJ, Thavaneswaran P, Coventry BJ (2014) Evaluation of surgical safety and efficacy. Springer, London
20. Margulis AV, Pladevall M, Riera-Guardia N, Varas-Lorenzo C, Hazell L, Berkman ND, Viswanathan M, Perez-Gutthann S (2014) Quality assessment of observational studies in a drug-safety systematic review, comparison of two tools: the Newcastle–Ottawa Scale and the RTI item bank. Clin Epidemiol 6(8):359–368
21. Cho P, Cheung SW (2017) Discontinuation of orthokeratology on eyeball elongation (DOEE). Cont Lens Anterior Eye 40(2):82–87. https://doi.org/10.1016/j.clae.2016.12.002
22. Chen C, Cheung SW, Cho P (2013) Myopia control using toric orthokeratology (TO-SEE study). Invest Ophthalmol Vis Sci 54(10):6510–6517. https://doi.org/10.1167/iovs.13-12527
23. Santodomingo-Rubido J, Villa-Collar C, Gilmartin B, Gutierrez-Ortega R, Sugimoto K (2017) Long-term efficacy of orthokeratology contact lens wear in controlling the progression of childhood myopia. Curr Eye Res 42(5):713–720. https://doi.org/10.1080/02713683.2016.1221979
24. Charm J, Cho P (2013) High myopia-partial reduction ortho-k: a 2-year randomized study. Optom Vis Sci 90(6):530–539. https://doi.org/10.1097/OPX.0b013e318293657d
25. He M, Du Y, Liu Q, Ren C, Liu J, Wang Q, Li L, Yu J (2016) Effects of orthokeratology on the progression of low to moderate myopia in Chinese children. BMC Ophthalmol 16:126. https://doi.org/10.1186/s12886-016-0302-5
26. Cho P, Cheung SW (2012) Retardation of myopia in Orthokeratology (ROMIO) study: a 2-year randomized clinical trial. Invest Ophthalmol Vis Sci 53(11):7077–7085. https://doi.org/10.1167/iovs.12-10565
27. Cheung SW, Cho P (2016) Long-term effect of orthokeratology on the anterior segment length. Cont Lens Anterior Eye 39(4):262–265. https://doi.org/10.1016/j.clae.2016.02.003
28. Kakita T, Hiraoka T, Oshika T (2011) Influence of overnight orthokeratology on axial elongation in childhood myopia. Invest Ophthalmol Vis Sci 52(5):2170–2174. https://doi.org/10.1167/iovs.10-5485
29. Paun J (2015) Myopia control with a novel peripheral gradient soft lens and orthokeratology: a 2-year clinical trial. Biomed Res Int 2015(D):507572. https://doi.org/10.1155/2015/507572
30. MaYA Na (2018) The effect of orthokeratology on axial length elongation in children with myopia: contralateral comparison study. Jpn J Ophthalmol 62(3):327–334. https://doi.org/10.1007/s10384-018-0573-x
31. Holden BA, Fricke TR, Wilson DA, Jong M, Naidoo KS, Sankaridurg P, Wong TY, Naduvilath TJ, Resnikoff S (2016) Global prevalence of myopia and high myopia and temporal trends from 2000 through 2050. Ophthalmology 123(5):1036–1042. https://doi.org/10.1016/j.ophtha.2016.01.006
32. Meng W, Butterworth J, Malecaze F, Calvas P (2011) Axial length of myopia: a review of current research. Ophthalmologica 225(3):127–134. https://doi.org/10.1159/000317072
33. Young TL, Metlapally R, Shay AE (2007) Complex trait genetics of refractive error. Arch Ophthalmol 125(1):38–48. https://doi.org/10.1001/archopht.125.1.38
34. Smith MJ, Walline JJ (2015) Controlling myopia progression in children and adolescents. Adolesc Health Med Ther 6:133–140. https://doi.org/10.2147/AHMT.S55834
35. Aller TA (2014) Clinical management of progressive myopia. Eye (Lond) 28(2):147–153. https://doi.org/10.1038/eye.2013.259
36. Cooper J, Schulman E, Jamal N (2012) Current status on the development and treatment of myopia. Optometry 83(5):179–199
37. Liu YM, Xie P (2016) The safety of orthokeratology—a systematic review. Eye Contact Lens 42(1):35–42. https://doi.org/10.1097/ICL.0000000000000219
38. Caroline PJ (2001) Contemporary orthokeratology. Cont Lens Anterior Eye 24(1):41–46

39. Lum E, Swarbrick HA (2011) Lens Dk/t influences the clinical response in overnight orthokeratology. Optom Vis Sci 88(4):469–475. https://doi.org/10.1097/OPX.0b013e31820bb0db
40. Cheung SW, Cho P, Fan D (2004) Asymmetrical increase in axial length in the two eyes of a monocular orthokeratology patient. Optom Vis Sci 81(9):653–656
41. Si JK, Tang K, Bi HS, Guo DD, Guo JG, Wang XR (2015) Orthokeratology for myopia control: a meta-analysis. Optom Vis Sci 92(3):252–257. https://doi.org/10.1097/OPX.0000000000000505
42. Cho P, Cheung SW, Edwards M (2005) The longitudinal orthokeratology research in children (LORIC) in Hong Kong: a pilot study on refractive changes and myopic control. Curr Eye Res 30(1):71–80
43. Santodomingo-Rubido J, Villa-Collar C, Gilmartin B, Gutierrez-Ortega R (2012) Myopia control with orthokeratology contact lenses in Spain: refractive and biometric changes. Invest Ophthalmol Vis Sci 53(8):5060–5065. https://doi.org/10.1167/iovs.11-8005

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